Juan Menduiña

Trabajo Práctico N° 5:
Ecuaciones en Diferencias.

Ejercicio 1.

Resolver las siguientes ecuaciones en diferencias lineales de primer orden:

(a) $\Delta y_t = 7$.

$y_t - y_{t-1} = 7$
$y_t = y_{t-1} + 7$.

$y_1 = y_0 + 7$.

$y_2 = y_1 + 7$
$y_2 = y_0 + 7 + 7$
$y_2 = y_0 + 14$.

$y_3 = y_2 + 7$
$y_3 = y_0 + 14 + 7$
$y_3 = y_0 + 21$.

$y_t = y_0 + 7t$.

(b) $\Delta y_t = 0,3 y_t$.

$y_t - y_{t-1} = 0,3 y_t$
$y_t - 0,3 y_t - y_{t-1} = 0$
$0,7 y_t - y_{t-1} = 0$
$0,7 y_t = y_{t-1}$
$y_t = \frac{1}{0,7} y_{t-1}$
$y_t = \frac{10}{7} y_{t-1}$.

$y_1 = \frac{10}{7} y_0$.

$y_2 = \frac{10}{7} y_1$
$y_2 = \frac{10}{7} \frac{10}{7} y_0$
$y_2 = (\frac{10}{7})^2 y_0$.

$y_3 = \frac{10}{7} y_2$
$y_3 = \frac{10}{7} (\frac{10}{7})^2 y_0$
$y_3 = (\frac{10}{7})^3 y_0$.

$y_t = (\frac{10}{7})^t\ y_0$.

(c) $\Delta y_t = 2y_t - 9$.

$y_t - y_{t-1} = 2y_t - 9$
$2y_t - y_t = -y_{t-1} + 9$
$y_t = -y_{t-1} + 9$.

$y_1 = -y_0 + 9$.

$y_2 = -y_1 + 9$
$y_2 = -(y_0 + 9) + 9$
$y_2 = -y_0 - 9 + 9$
$y_2 = y_0$.

$y_3 = -y_2 + 9$
$y_3 = -y_0 + 9$.

$$y_t = \begin{cases} y_0, t\ par \\ -y_0 + 9, t\ impar \end{cases}.$$

Ejercicio 2.

Resolver las siguientes ecuaciones en diferencias lineales de primer orden con condición inicial y_0:

(a) $\begin{cases} y_t = y_{t-1} + 1 \\ y_0 = 10 \end{cases}$.

$y_1 = y_0 + 1$.

$y_2 = y_1 + 1$
$y_2 = y_0 + 1 + 1$
$y_2 = y_0 + 2$.

$y_3 = y_2 + 1$
$y_3 = y_0 + 2 + 1$
$y_3 = y_0 + 3$.

$y_t = y_0 + t$.

$y_t = 10 + t$.

(b) $\begin{cases} y_t + 3y_{t-1} = 4 \\ y_0 = 4 \end{cases}$.

$y_t = -3y_{t-1} + 4$.

$y_1 = -3y_0 + 4$
$y_1 = -3 * 4 + 4$
$y_1 = -12 + 4$
$y_1 = -8$.

$y_2 = -3y_1 + 4$
$y_2 = -3 (-3y_0 + 4) + 4$
$y_2 = 9y_0 - 12 + 4$
$y_2 = 9y_0 - 8$
$y_2 = 9 * 4 - 8$
$y_2 = 36 - 8$
$y_2 = -28$.

$y_3 = -3y_2 + 4$
$y_3 = -3 (9y_0 - 8) + 4$
$y_3 = -27y_0 + 24 + 4$
$y_3 = -27y_0 + 28$
$y_3 = -27 * 4 + 28$
$y_3 = -108 + 28$
$y_3 = -80$.

c + 3c= 4
4c= 4
c= $\frac{4}{4}$
c= 1.

y_t - c= (y_0 - c) $(-3)^t$
y_t - 1= (4 - 1) $(-3)^t$
y_t - 1= 3 $(-3)^t$
y_t= 1 + 3 $(-3)^t$.

(c) $\begin{cases} 2y_t - y_{t-1} = 6 \\ y_0 = 7 \end{cases}$.

$2y_t$= y_{t-1} + 6
y_t= $\frac{y_{t-1}+6}{2}$
y_t= $\frac{1}{2}$ y_{t-1} + 3.

y_1= $\frac{1}{2}$ y_0 + 3
y_1= $\frac{1}{2}$ * 7 + 3
y_1= $\frac{7}{2}$ + 3
y_1= $\frac{13}{2}$.

y_2= $\frac{1}{2}$ y_1 + 3
y_2= $\frac{1}{2}$ ($\frac{1}{2}$ y_0 + 3) + 3
y_2= $\frac{1}{4}$ y_0 + $\frac{3}{2}$ + 3
y_2= $\frac{1}{4}$ y_0 + $\frac{9}{2}$
y_2= $\frac{1}{4}$ * 7 + $\frac{9}{2}$
y_2= $\frac{7}{4}$ + $\frac{9}{2}$
y_2= $\frac{25}{4}$.

y_3= $\frac{1}{2}$ y_2 + 3
y_3= $\frac{1}{2}$ ($\frac{1}{4}$ y_0 + $\frac{9}{2}$) + 3
y_3= $\frac{1}{8}$ y_0 + $\frac{9}{4}$ + 3
y_3= $\frac{1}{8}$ y_0 + $\frac{21}{4}$
y_3= $\frac{1}{8}$ * 7 + $\frac{21}{4}$
y_3= $\frac{7}{8}$ + $\frac{21}{4}$
y_3= $\frac{49}{8}$.

2c - c= 6
c= 6.

y_t - c= (y_0 - c) $(\frac{1}{2})^t$
y_t - 6= (7 - 6) $(\frac{1}{2})^t$
y_t - 6= $(\frac{1}{2})^t$
y_t= 6 + $(\frac{1}{2})^t$.

(d) $\begin{cases} y_t = 0{,}2y_{t-1} + 4 \\ y_0 = 4 \end{cases}$.

y_1= 0,2y_0 + 4
y_1= 0,2 * 4 + 4
y_1= 0,8 + 4
y_1= 4,8.

y_2= 0,2y_1 + 4
y_2= 0,2 (0,2y_0 + 4) + 4
y_2= 0,04y_0 + 0,8 + 4
y_2= 0,04y_0 + 4,8
y_2= 0,04 * 4 + 4,8
y_2= 0,16 + 4,8
y_2= 4,96.

y_3= 0,2y_2 + 4
y_3= 0,2 (0,04y_0 + 4,8) + 4
y_3= 0,008y_0 + 0,96 + 4
y_3= 0,008y_0 + 4,96
y_3= 0,008 * 4 + 4,96
y_3= 0,032 + 4,96
y_3= 4,992.

c= 0,2c + 4
c - 0,2= 4
$\frac{4}{5}$ c= 4
c= $\frac{4}{\frac{4}{5}}$
c= 5.

y_t - c= (y_0 - c) $(0{,}2)^t$
y_t - 5= (4 - 5) $(0{,}2)^t$
y_t - 5= -$(0{,}2)^t$
y_t= 5 - $0{,}2^t$.

Ejercicio 3.

Encontrar las soluciones de las siguientes ecuaciones en diferencias y determinar si las soluciones convergen u oscilan en t= ∞:

(a) $\begin{cases} y_t - \frac{1}{3} y_{t-1} = 6 \\ y_0 = 1 \end{cases}$.

$y_t = \frac{1}{3} y_{t-1} + 6.$

$y_1 = \frac{1}{3} y_0 + 6$
$y_1 = \frac{1}{3} * 1 + 6$
$y_1 = \frac{1}{3} + 6$
$y_1 = \frac{19}{3}.$

$y_2 = \frac{1}{3} y_1 + 6$
$y_2 = \frac{1}{3} (\frac{1}{3} y_0 + 6) + 6$
$y_2 = \frac{1}{9} y_0 + 2 + 6$
$y_2 = \frac{1}{9} y_0 + 8$
$y_2 = \frac{1}{9} * 1 + 8$
$y_2 = \frac{1}{9} + 8$
$y_2 = \frac{73}{9}.$

$y_3 = \frac{1}{3} y_2 + 6$
$y_3 = \frac{1}{3} (\frac{1}{9} y_0 + 8) + 6$
$y_3 = \frac{1}{27} y_0 + \frac{8}{3} + 6$
$y_3 = \frac{1}{27} y_0 + \frac{26}{3}$
$y_3 = \frac{1}{27} * 1 + \frac{26}{3}$
$y_3 = \frac{1}{27} + \frac{26}{3}$
$y_3 = \frac{235}{27}.$

c - $\frac{1}{3}$ c= 6
$\frac{2}{3}$ c= 6
c= $\frac{6}{\frac{2}{3}}$
c= 9.

y_t - c= (y_0 - c) $(\frac{1}{3})^t$
y_t - 9= (1 - 9) $(\frac{1}{3})^t$

y_t - 9= -8 $(\frac{1}{3})^t$
y_t= 9 - 8 $(\frac{1}{3})^t$.

$\lim\limits_{t \to +\infty} y_t$= 9.

Por lo tanto, la solución converge a 9 y no oscila, ya que $\left|\frac{1}{3}\right| < 1$ y $\frac{1}{3} > 0$.

(b) $\begin{cases} y_t + 2y_{t-1} = 9 \\ y_0 = 4 \end{cases}$.

y_t= -2y_{t-1} + 9.

y_1= -2y_0 + 9
y_1= -2 * 4 + 9
y_1= -8 + 9
y_1= 1.

y_2= -2y_1 + 9
y_2= -2 (-2y_0 + 9) + 9
y_2= 4y_0 - 18 + 9
y_2= 4y_0 - 9
y_2= 4 * 4 - 9
y_2= 16 - 9
y_2= 7.

y_3= -2y_2 + 9
y_3= -2 (4y_0 - 9) + 9
y_3= -8y_0 + 18 + 9
y_3= -8y_0 + 27
y_3= -8 * 4 + 27
y_3= -32 + 27
y_3= -5.

c + 2c= 9
3c= 9
c= $\frac{9}{3}$
c= 3.

y_t - c= (y_0 - c) $(-2)^t$
y_t - 3= (4 - 3) $(-2)^t$
y_t - 3= $(-2)^t$
y_t= 3 + $(-2)^t$.

Por lo tanto, la solución no converge y oscila, ya que $|-2| > 1$ y -2 < 0.

(c) $\begin{cases} 2y_t + \frac{1}{4}y_{t-1} = 5 \\ y_0 = 2 \end{cases}$.

$2y_t = \frac{-1}{4}\, y_{t-1} + 5$
$y_t = \frac{\frac{-1}{4}y_{t-1}+5}{2}$
$y_t = \frac{-1}{8}\, y_{t-1} + \frac{5}{2}$.

$y_1 = \frac{-1}{8}\, y_0 + \frac{5}{2}$
$y_1 = \frac{-1}{8} * 2 + \frac{5}{2}$
$y_1 = \frac{-1}{4} + \frac{5}{2}$
$y_1 = \frac{9}{4}$.

$y_2 = \frac{-1}{8}\, y_1 + \frac{5}{2}$
$y_2 = \frac{-1}{8}\, (\frac{-1}{8}\, y_0 + \frac{5}{2}) + \frac{5}{2}$
$y_2 = \frac{1}{64}\, y_0 - \frac{5}{16} + \frac{5}{2}$
$y_2 = \frac{1}{64}\, y_0 + \frac{35}{16}$
$y_2 = \frac{1}{64} * 2 + \frac{35}{16}$
$y_2 = \frac{1}{32} + \frac{35}{16}$
$y_2 = \frac{71}{32}$.

$y_3 = \frac{-1}{8}\, y_2 + \frac{5}{2}$
$y_3 = \frac{-1}{8}\, (\frac{1}{64}\, y_0 + \frac{35}{16}) + \frac{5}{2}$
$y_3 = \frac{-1}{512}\, y_0 - \frac{35}{128} + \frac{5}{2}$
$y_3 = \frac{-1}{512}\, y_0 + \frac{285}{128}$
$y_3 = \frac{-1}{512} * 2 + \frac{285}{128}$
$y_3 = \frac{-1}{256} + \frac{285}{128}$
$y_3 = \frac{569}{256}$.

$2c + \frac{1}{4}\, c = 5$
$\frac{9}{4}\, c = 5$
$c = \frac{5}{\frac{9}{4}}$
$c = \frac{20}{9}$.

$y_t - c = (y_0 - c)\, (\frac{-1}{8})^t$
$y_t - \frac{20}{9} = (2 - \frac{20}{9})\, (\frac{-1}{8})^t$

y_t - $\frac{20}{9}$= $\frac{-2}{9}$ $(\frac{-1}{8})^t$
y_t= $\frac{20}{9}$ - $\frac{2}{9}$ $(\frac{-1}{8})^t$.

$\lim_{t \to +\infty} y_t = \frac{20}{9}$.

Por lo tanto, la solución converge a $\frac{20}{9}$ y oscila, ya que $\left|\frac{-1}{8}\right| < 1$ y $\frac{-1}{8} < 0$.

(d) $\begin{cases} y_t - y_{t-1} = 3 \\ y_0 = 5 \end{cases}$.

y_t= y_{t-1} + 3.

y_1= y_0 + 3
y_1= 5 + 3
y_1= 8.

y_2= y_1 + 3
y_2= y_0 + 3 + 3
y_2= y_0 + 6
y_2= 5 + 6
y_2= 11.

y_3= y_2 + 3
y_3= y_0 + 6 + 3
y_3= y_0 + 9
y_3= 5 + 9
y_3= 14.

y_t= 5 + 3t.

$\lim_{t \to +\infty} y_t = +\infty$.

Por lo tanto, la solución no converge y no oscila (diverge hacia +∞), ya que $|3| > 1$ y $3 > 0$.

Ejercicio 4.

Encontrar todas las soluciones de las siguientes ecuaciones:

(a) y_{t+2} - y_{t+1} + $\frac{1}{2}$ y_t= *2.*

Solución homogénea:

y_t= r^t.

r^{t+2} - r^{t+1} + $\frac{1}{2}$ r^t= 0
r^t (r^2 - r + $\frac{1}{2}$)= 0.

r_1, r_2= $\frac{-(-1)\pm\sqrt{(-1)^2-4*1*\frac{1}{2}}}{2*1}$
r_1, r_2= $\frac{1\pm\sqrt{1-2}}{2}$
r_1, r_2= $\frac{1\pm\sqrt{-1}}{2}$
r_1, r_2= $\frac{1\pm i}{2}$
r_1= $\frac{1+i}{2}$= $\frac{1}{2}$ + $\frac{1}{2}$ i.
r_2= $\frac{1-i}{2}$= $\frac{1}{2}$ - $\frac{1}{2}$ i.

ρ= $\sqrt{(\frac{1}{2})^2 + (\frac{1}{2})^2}$
ρ= $\sqrt{\frac{1}{4} + \frac{1}{4}}$
ρ= $\sqrt{\frac{1}{2}}$
ρ= $\frac{1}{\sqrt{2}}$.

tan θ= $\frac{\frac{1}{2}}{\frac{1}{2}}$
tan θ= 1.

r_1, r_2= $\frac{1}{\sqrt{2}}$ (cos $\frac{\pi}{4}$ $\pm$ i sen $\frac{\pi}{4}$).

y_t^h= $(\frac{1}{\sqrt{2}})^t$ (C_1 cos $\frac{\pi t}{4}$ + C_2 sen $\frac{\pi t}{4}$).

Solución particular:

y_t^p= k.

k - k + $\frac{1}{2}$ k= 2

$\frac{1}{2}$ k= 2
k= 2 * 2
k= 4.

y_t^p= 4.

<u>Solución general:</u>

y_t= y_t^h + y_t^p
y_t= $(\frac{1}{\sqrt{2}})^t$ (C_1 cos $\frac{\pi t}{4}$ + C_2 sen $\frac{\pi t}{4}$) + 4.

(b) y_{t+2} - $4y_{t+1}$ + $4y_t$= 7.

<u>Solución homogénea:</u>

y_t= r^t.

r^{t+2} - 4r^{t+1} + 4r^t= 0
r^t (r^2 - 4r + 4)= 0.

r_1, r_2= $\frac{-(-4)\pm\sqrt{(-4)^2-4*1*4}}{2*1}$
r_1, r_2= $\frac{4\pm\sqrt{16-16}}{2}$
r_1, r_2= $\frac{4\pm\sqrt{0}}{2}$
r_1, r_2= $\frac{4\pm 0}{2}$
r_1= $\frac{4+0}{2}$= $\frac{4}{2}$= 2.
r_2= $\frac{4-0}{2}$= $\frac{4}{2}$= 2.

y_t^h= (C_1 + C_2t) 2^t.

<u>Solución particular:</u>

y_t^p= k.

k - 4k + 4k= 7
k= 7.

y_t^p= 7.

<u>Solución general:</u>

y_t= y_t^h + y_t^p
y_t= (C_1 + C_2t) 2^t + 7.

(c) $2y_{t+2} + y_{t+1} - y_t = 10$.

Solución homogénea:

$y_t = r^t$.

$2r^{t+2} + r^{t+1} - r^t = 0$
$r^t\ (2r^2 + r - 1) = 0$.

$r_1, r_2 = \frac{-1 \pm \sqrt{1^2 - 4*2(-1)}}{2*2}$
$r_1, r_2 = \frac{-1 \pm \sqrt{1+8}}{4}$
$r_1, r_2 = \frac{-1 \pm \sqrt{9}}{4}$
$r_1, r_2 = \frac{-1 \pm 3}{4}$
$r_1 = \frac{-1+3}{4} = \frac{2}{4} = \frac{1}{2}$.
$r_2 = \frac{-1-3}{4} = \frac{-2}{4} = -1$.

$y_t^h = C_1\ (\frac{1}{2})^t + C_2\ (-1)^t$.

Solución particular:

$y_t^p = k$.

$2k + k - k = 10$
$2k = 10$
$k = \frac{10}{2}$
$k = 5$.

$y_t^p = 5$.

Solución general:

$y_t = y_t^h + y_t^p$
$y_t = C_1\ (\frac{1}{2})^t + C_2\ (-1)^t + 5$.

(d) $y_{t+2} - 2y_{t+1} + 3y_t = 4$.

Solución homogénea:

$y_t = r^t$.

$r^{t+2} - 2r^{t+1} + 3r^t = 0$

r^t (r^2 - 2r + 3)= 0.

$r_1, r_2 = \frac{-(-2) \pm \sqrt{(-2)^2 - 4*1*3}}{2*1}$
$r_1, r_2 = \frac{4 \pm \sqrt{4-12}}{2}$
$r_1, r_2 = \frac{1 \pm \sqrt{-8}}{2}$
$r_1, r_2 = \frac{1 \pm \sqrt{4*2(-1)}}{2}$
$r_1, r_2 = \frac{1 \pm 2\sqrt{2}i}{2}$
$r_1 = \frac{1+2\sqrt{2}i}{2} = \frac{1}{2} + \sqrt{2}$ i.
$r_2 = \frac{1-2\sqrt{2}i}{2} = \frac{1}{2} - \sqrt{2}$ i.

$\rho = \sqrt{1^2 + (\sqrt{2})^2}$
$\rho = \sqrt{1+2}$
$\rho = \sqrt{3}$.

tan $\theta = \frac{\sqrt{2}}{1}$
tan $\theta = \sqrt{2}$.

$r_1, r_2 = \sqrt{3}$ (cos (arctan $\sqrt{2}$) $\pm$ i sen (arctan $\sqrt{2}$)).

$y_t^h = (\sqrt{3})^t$ [C_1 cos (t arctan $\sqrt{2}$) + C_2 sen (t arctan $\sqrt{2}$)].

<u>Solución particular:</u>

y_t^p= k.

k - 2k + 3k= 4
2k= 4
k= $\frac{4}{2}$
k= 2.

y_t^p= 2.

<u>Solución general:</u>

$y_t = y_t^h + y_t^p$
$y_t = (\sqrt{3})^t$ [C_1 cos (t arctan $\sqrt{2}$) + C_2 sen (t arctan $\sqrt{2}$)] + 2.

Ejercicio 5.

Encontrar las soluciones de las siguientes ecuaciones en diferencias y determinar si las soluciones convergen u oscilan en t= ∞:

(a) $\begin{cases} y_{t+2} + 3y_{t+1} - \frac{7}{4}y_t = 9 \\ y_0 = 6 \\ y_1 = 3 \end{cases}$.

Solución homogénea:

$y_t = r^t$.

$r^{t+2} + 3r^{t+1} - \frac{7}{4} r^t = 0$
$r^t\ (r^2 + 3r - \frac{7}{4}) = 0$.

$r_1, r_2 = \frac{-3 \pm \sqrt{3^2 - 4*1(\frac{-7}{4})}}{2*1}$
$r_1, r_2 = \frac{-3 \pm \sqrt{9+7}}{2}$
$r_1, r_2 = \frac{-3 \pm \sqrt{16}}{2}$
$r_1, r_2 = \frac{-3 \pm 4}{2}$
$r_1 = \frac{-3+4}{2} = \frac{1}{2}$.
$r_2 = \frac{-3-4}{2} = \frac{-7}{2}$.

$y_t^h = C_1\ (\frac{1}{2})^t + C_2\ (\frac{-7}{2})^t$.

Solución particular:

$y_t^p = k$.

$k + 3k - \frac{7}{4} k = 9$
$\frac{9}{4} = 9$
$k = \frac{9}{\frac{9}{4}}$
$k = 4$.

$y_t^p = 4$.

Solución general:

$y_t = y_t^h + y_t^p$
$y_t = C_1\ (\frac{1}{2})^t + C_2\ (\frac{-7}{2})^t + 4$.

Solución final:

$y_0 = C_1\ (\frac{1}{2})^0 + C_2\ (\frac{-7}{2})^0 + 4$
$y_0 = C_1 * 1 + C_2 * 1 + 4$
$y_0 = C_1 + C_2 + 4$
$6 = C_1 + C_2 + 4$
$C_1 + C_2 = 6 - 4$
$C_1 + C_2 = 2.$

$y_1 = C_1\ (\frac{1}{2})^1 + C_2\ (\frac{-7}{2})^1 + 4$
$y_1 = C_1\ \frac{1}{2} + C_2\ (\frac{-7}{2}) + 4$
$3 = \frac{1}{2}\ C_1 - \frac{7}{2}\ C_2 + 4$
$\frac{1}{2}\ C_1 - \frac{7}{2}\ C_2 = 3 - 4$
$\frac{1}{2}\ C_1 - \frac{7}{2}\ C_2 = -1$
$\frac{C_1 - 7C_2}{2} = -1$
$C_1 - 7C_2 = -1 * 2$
$C_1 - 7C_2 = -2.$

$\begin{cases} C_1 + C_2 = 2 \\ C_1 - 7C_2 = -2 \end{cases}.$

$C_2 = 2 - C_1.$

$C_1 - 7\ (2 - C_1) = -2$
$C_1 - 14 + 7C_1 = -2$
$8C_1 = -2 + 14$
$8C_1 = 12$
$C_1 = \frac{12}{8}$
$C_1 = \frac{3}{2}.$

$C_2 = 2 - \frac{3}{2}$
$C_2 = \frac{1}{2}.$

$y_t = \frac{3}{2}\ (\frac{1}{2})^t + \frac{1}{2}\ (\frac{-7}{2})^t + 4.$

Por lo tanto, la solución no converge y oscila, ya que $\left|\frac{-7}{2}\right| > 1$ y $\frac{-7}{2} < 0$.

(b) $\begin{cases} y_{t+2} - 2y_{t+1} + 2y_t = 1 \\ y_0 = 3 \\ y_1 = 4 \end{cases}.$

Solución homogénea:

$y_t = r^t$.

r^{t+2} - $2r^{t+1}$ + $2r^t$= 0
r^t (r^2 - 2r + 2)= 0.

$r_1, r_2 = \frac{-(-2) \pm \sqrt{(-2)^2 - 4*1*2}}{2*1}$
$r_1, r_2 = \frac{2 \pm \sqrt{4-8}}{2}$
$r_1, r_2 = \frac{2 \pm \sqrt{-4}}{2}$
$r_1, r_2 = \frac{2 \pm 2\sqrt{-1}}{2}$
$r_1, r_2 = \frac{2 \pm 2i}{2}$
$r_1 = \frac{2+2i}{2}$= 1 + i.
$r_2 = \frac{2-2i}{2}$= 1 - i.

$\rho = \sqrt{1^2 + 1^2}$
$\rho = \sqrt{1+1}$
$\rho = \sqrt{2}$.

tan $\theta = \frac{1}{1}$
tan θ= 1.

$r_1, r_2 = \sqrt{2}$ (cos $\frac{\pi}{4} \pm$ i sen $\frac{\pi}{4}$).

$y_t^h = (\sqrt{2})^t$ (C_1 cos $\frac{\pi t}{4}$ + C_2 sen $\frac{\pi t}{4}$).

<u>Solución particular:</u>

y_t^p= k.

k - 2k + 2k= 1
k= 1.

y_t^p= 1.

<u>Solución general:</u>

$y_t = y_t^h + y_t^p$
$y_t = (\sqrt{2})^t$ (C_1 cos $\frac{\pi t}{4}$ + C_2 sen $\frac{\pi t}{4}$) + 1.

<u>Solución final:</u>

$y_0 = (\sqrt{2})^0$ (C_1 cos $\frac{\pi * 0}{4}$ + C_2 sen $\frac{\pi * 0}{4}$) + 1
y_0= 1 (C_1 cos 0 + C_2 sen 0) + 1

y_0= (C_1 * 1 + C_2 * 0) + 1
y_0= (C_1 + 0) + 1
3= C_1 + 1
C_1= 3 - 1
C_1= 2.

y_1= $(\sqrt{2})^1$ (2 cos $\frac{\pi*1}{4}$ + C_2 sen $\frac{\pi*1}{4}$) + 1
y_1= $\sqrt{2}$ (2 cos $\frac{\pi}{4}$ + C_2 sen $\frac{\pi}{4}$) + 1
y_1= $\sqrt{2}$ (2 $\frac{\sqrt{2}}{2}$ + C_2 $\frac{\sqrt{2}}{2}$) + 1
y_1= $\sqrt{2}$ ($\sqrt{2}$ + C_2 $\frac{\sqrt{2}}{2}$) + 1
y_1= (2 + C_2 $\frac{2}{2}$) + 1
y_1= (2 + C_2 * 1) + 1
y_1= (2 + C_2) + 1
y_1= 2 + C_2 + 1
y_1= 3 + C_2
4= 3 + C_2
C_2= 4 - 3
C_2= 1.

y_t= $(\sqrt{2})^t$ (2 cos $\frac{\pi t}{4}$ + sen $\frac{\pi t}{4}$) + 1.

Por lo tanto, la solución no converge y oscila (por el seno y coseno), ya que $|\sqrt{2}| > 1$.

(c) $\begin{cases} y_{t+2} - y_{t+1} + \frac{1}{4} y_t = 2 \\ y_0 = 4 \\ y_1 = 7 \end{cases}$.

<u>Solución homogénea:</u>

y_t= r^t.

r^{t+2} - r^{t+1} + $\frac{1}{4}$ r^t= 0
r^t (r^2 - r + $\frac{1}{4}$)= 0.

r_1, r_2= $\frac{-(-1)\pm\sqrt{(-1)^2-4*1\frac{1}{4}}}{2*1}$
r_1, r_2= $\frac{1\pm\sqrt{1-1}}{2}$
r_1, r_2= $\frac{1\pm\sqrt{0}}{2}$
r_1, r_2= $\frac{1\pm0}{2}$
r_1, r_2= $\frac{1+0}{2}$= $\frac{1}{2}$.

$r_1, r_2 = \frac{1-0}{2} = \frac{1}{2}$.

$y_t^h = (C_1 + C_2 t)\ (\frac{1}{2})^t$.

Solución particular:

$y_t^p = k$.

$k - k + \frac{1}{4} k = 2$
$\frac{1}{4} k = 2$
$k = 2 * 4$
$k = 8$.

$y_t^p = 8$.

Solución general:

$y_t = y_t^h + y_t^p$
$y_t = (C_1 + C_2 t)\ (\frac{1}{2})^t + 8$.

Solución final:

$y_0 = (C_1 + C_2 * 0)\ (\frac{1}{2})^0 + 8$
$y_0 = (C_1 + 0) * 1 + 8$
$y_0 = C_1 * 1 + 8$
$y_0 = C_1 + 8$
$4 = C_1 + 8$
$C_1 = 4 - 8$
$C_1 = -4$.

$y_1 = (-4 + C_2 * 1)\ (\frac{1}{2})^1 + 8$
$y_1 = (-4 + C_2)\ \frac{1}{2} + 8$
$y_1 = -2 + \frac{1}{2} C_2 + 8$
$y_1 = \frac{1}{2} C_2 + 6$
$7 = \frac{1}{2} C_2 + 6$
$\frac{1}{2} C_2 = 7 - 6$
$\frac{1}{2} C_2 = 1$
$C_2 = 1 * 2$
$C_2 = 2$.

$y_t = (-4 + 2t)\ (\frac{1}{2})^t + 8$.

Por lo tanto, la solución converge a 8 y no oscila, ya que $\left|\frac{1}{2}\right| < 1$ y $\frac{1}{2} > 0$.

(d) $\begin{cases} 2y_{t+2} + 2y_{t+1} + y_t = 2^{-t} \\ y_0 = 0 \\ y_1 = 0 \end{cases}$.

<u>Solución homogénea:</u>

$y_t = r^t$.

$2r^{t+2} + 2r^{t+1} + r^t = 0$
$r^t\ (2r^2 + 2r + 1) = 0$.

$r_1, r_2 = \frac{-2 \pm \sqrt{2^2 - 4*1*2}}{2*2}$
$r_1, r_2 = \frac{-2 \pm \sqrt{4-8}}{4}$
$r_1, r_2 = \frac{-2 \pm \sqrt{-4}}{4}$
$r_1, r_2 = \frac{-2 \pm 2\sqrt{-1}}{4}$
$r_1, r_2 = \frac{-2 \pm 2i}{4}$
$r_1 = \frac{-2+2i}{4} = \frac{-1}{2} + \frac{1}{2}$ i.
$r_2 = \frac{-2-2i}{4} = \frac{-1}{2} - \frac{1}{2}$ i.

$\rho = \sqrt{(\frac{-1}{2})^2 + (\frac{1}{2})^2}$
$\rho = \sqrt{\frac{1}{4} + \frac{1}{4}}$
$\rho = \sqrt{\frac{1}{2}}$
$\rho = \frac{1}{\sqrt{2}}$.

$\tan \theta = \frac{\frac{1}{2}}{\frac{-1}{2}}$
$\tan \theta = -1$.

$r_1, r_2 = \frac{1}{\sqrt{2}} (\cos \frac{3\pi}{4} \pm \text{i sen} \frac{3\pi}{4})$.

$y_t^h = (\frac{1}{\sqrt{2}})^t\ (C_1 \cos \frac{3\pi t}{4} + C_2 \text{ sen} \frac{3\pi t}{4})$.

<u>Solución particular:</u>

$y_t^p = k2^{-t}$.

$y_{t+1} = k\ 2^{-(t+1)}$

y_{t+1}= k 2^{-t-1}
y_{t+1}= $\frac{k}{2}$ 2^{-t}.

y_{t+2}= k $2^{-(t+2)}$
y_{t+2}= k 2^{-t-2}
y_{t+2}= $\frac{k}{4}$ 2^{-t}.

2 $\frac{k}{4}$ 2^{-t} + 2 $\frac{k}{2}$ 2^{-t} + k2^{-t}= 2^{-t}
$\frac{k}{2}$ 2^{-t} + k2^{-t} + k2^{-t}= 2^{-t}
($\frac{k}{2}$ + k + k) 2^{-t}= 2^{-t}
$\frac{5}{2}$ k 2^{-t}= 2^{-t}
$\frac{5}{2}$ k= $\frac{2^{-t}}{2^{-t}}$
$\frac{5}{2}$ k= 1
k= $\frac{1}{\frac{5}{2}}$
k= $\frac{2}{5}$.

y_t^p= $\frac{2}{5}$ 2^{-t}.

<u>Solución general:</u>

y_t= y_t^h + y_t^p
y_t= $(\frac{1}{\sqrt{2}})^t$ (C_1 cos $\frac{3\pi t}{4}$ + C_2 sen $\frac{3\pi t}{4}$) + $\frac{2}{5}$ 2^{-t}.

<u>Solución final:</u>

y_0= $(\frac{1}{\sqrt{2}})^0$ (C_1 cos $\frac{3\pi*0}{4}$ + C_2 sen $\frac{3\pi*0}{4}$) + $\frac{2}{5}$ 2^{-0}
y_0= 1 (C_1 cos $\frac{0}{4}$ + C_2 sen $\frac{0}{4}$) + $\frac{2}{5}$ * 1
y_0= (C_1 cos 0 + C_2 sen 0) + $\frac{2}{5}$
y_0= (C_1 * 1 + C_2 * 0) + $\frac{2}{5}$
y_0= (C_1 + 0) + $\frac{2}{5}$
0= C_1 + $\frac{2}{5}$
C_1= $\frac{-2}{5}$.

y_1= $(\frac{1}{\sqrt{2}})^1$ ($\frac{-2}{5}$ cos $\frac{3\pi*1}{4}$ + C_2 sen $\frac{3\pi*1}{4}$) + $\frac{2}{5}$ 2^{-1}
y_1= $\frac{1}{\sqrt{2}}$ [$\frac{-2}{5}$ ($\frac{-\sqrt{2}}{2}$) + C_2 $\frac{\sqrt{2}}{2}$] + $\frac{1}{5}$
y_1= $\frac{1}{\sqrt{2}}$ ($\frac{\sqrt{2}}{5}$ + C_2 $\frac{\sqrt{2}}{2}$) + $\frac{1}{5}$
y_1= $\frac{1}{5}$ + $\frac{1}{2}$ C_2 + $\frac{1}{5}$

$y_1 = \frac{1}{2} C_2 + \frac{2}{5}$
$0 = \frac{1}{2} C_2 + \frac{2}{5}$
$\frac{1}{2} C_2 = \frac{-2}{5}$
$C_2 = \frac{\frac{-2}{5}}{\frac{1}{2}}$
$C_2 = \frac{-4}{5}$.

$y_t = (\frac{1}{\sqrt{2}})^t \, [\frac{-2}{5} \cos \frac{3\pi t}{4} + (\frac{-4}{5}) \text{ sen } \frac{3\pi t}{4}] + \frac{2}{5} \, 2^{-t}$.

$\lim_{t \to +\infty} y_t = 0$.

Por lo tanto, la solución converge a 0 y oscila (por el seno y coseno), ya que $\left|\frac{1}{\sqrt{2}}\right| < 1$ y $\left|\frac{1}{2}\right| < 1$.

Ejercicio 6.

Encontrar todas las soluciones de las siguientes ecuaciones:

(a) y_{t+2} + $2y_{t+1}$ + y_t= 3^t.

Solución homogénea:

y_t= r^t.

r^{t+2} + $2r^{t+1}$ + r^t= 0
r^t (r^2 + 2r + r)= 0.

r_1, r_2= $\frac{-2\pm\sqrt{2^2-4*1*1}}{2*1}$
r_1, r_2= $\frac{-2\pm\sqrt{4-4}}{2}$
r_1, r_2= $\frac{-2\pm\sqrt{0}}{2}$
r_1, r_2= $\frac{-2\pm0}{2}$
r_1, r_2= $\frac{-2+0}{2}$= $\frac{-2}{2}$= -1.
r_1, r_2= $\frac{-2-0}{2}$= $\frac{-2}{2}$= -1.

y_t^h= (C_1 + C_2t) $(-1)^t$.

Solución particular:

y_t^p= k3^t.

y_{t+1}= k 3^{t+1}
y_{t+1}= 3k 3^t.

y_{t+2}= k 3^{t+2}
y_{t+2}= 9k 3^t.

9k 3^t + 2 * 3k 3^t + k3^t= 3^t
9k 3^t + 6k 3^t + k3^t= 3^t
16k 3^t= 3^t
16k= $\frac{3^t}{3^t}$
16k= 1
k= $\frac{1}{16}$.

y_t^p= $\frac{3^t}{16}$.

Solución general:

$y_t = y_t^h + y_t^p$
$y_t = (C_1 + C_2 t)\,(-1)^t + \frac{3^t}{16}$.

(b) y_{t+2} - $5y_{t+1}$ - $6y_t$= $2 * 3^t$.

<u>Solución homogénea:</u>

$y_t = r^t$.

r^{t+2} - $5r^{t+1}$ - $6r^t$= 0
r^t (r^2 - 5r - 6)= 0.

$r_1, r_2 = \frac{-(-5)\pm\sqrt{(-5)^2-4*1(-6)}}{2*1}$
$r_1, r_2 = \frac{5\pm\sqrt{25+24}}{2}$
$r_1, r_2 = \frac{5\pm\sqrt{49}}{2}$
$r_1, r_2 = \frac{5\pm 7}{2}$
$r_1, r_2 = \frac{5+7}{2} = \frac{12}{2} = 6$.
$r_1, r_2 = \frac{5-7}{2} = \frac{-2}{2} = -1$.

$y_t^h = C_1\, 6^t + C_2\, (-1)^t$.

<u>Solución particular:</u>

y_t^p= k3^t.

y_{t+1}= k 3^{t+1}
y_{t+1}= 3k 3^t.

y_{t+2}= k 3^{t+2}
y_{t+2}= 9k 3^t.

9k 3^t - 5 * 3k 3^t - 6k 3^t= 2 * 3^t
9k 3^t - 15k 3^t - 6k 3^t= 2 * 3^t
-12k 3^t= 2 * 3^t
-12k= 2 $\frac{3^t}{3^t}$
k= $\frac{2}{-12}$
k= $\frac{-1}{6}$.

$y_t^p = \frac{-1}{6}\, 3^t$.

<u>Solución general:</u>

$y_t = y_t^h + y_t^p$
$y_t = C_1\ 6^t + C_2\ (-1)^t - \frac{1}{6}\ 3^t$.

(c) $3y_{t+2} + 9y_t = 3 * 4^t$.

Solución homogénea:

$y_t = r^t$.

$3r^{t+2} + 9r^t = 0$
$3r^t\ (r^2 + 3) = 0$.

$r^2 + 3 = 0$
$r^2 = -3$
$\sqrt{r} = \sqrt{-3}$
$|r| = \sqrt{-1}\ \sqrt{-3}$
$r = \pm\ \sqrt{3}i$.

$\rho = \sqrt{(\sqrt{3})^2}$
$\rho = \sqrt{3}$.

$\tan\theta = \frac{\sqrt{3}}{0}$.

$r_1, r_2 = \sqrt{3}\ (\cos\frac{\pi}{2} \pm i \operatorname{sen}\frac{\pi}{2})$.

$y_t^h = (\sqrt{3})^t\ (C_1 \cos\frac{\pi t}{2} + C_2 \operatorname{sen}\frac{\pi t}{2})$.

Solución particular:

$y_t^p = k4^t$.

$y_{t+1} = k\ 4^{t+1}$
$y_{t+1} = 4k\ 4^t$.

$y_{t+2} = k\ 4^{t+2}$
$y_{t+2} = 16k\ 4^t$.

$3 * 16k\ 4^t + 9k\ 4^t = 3 * 4^t$
$16k\ 4^t + 3k\ 4^t = 4^t$
$19k\ 4^t = 4^t$
$19k = \frac{4^t}{4^t}$
$19k = 1$
$k = \frac{1}{19}$,

$y_t^p = \frac{4^t}{19}$.

Solución general:

$y_t = y_t^h + y_t^p$
$y_t = (\sqrt{3})^t \; (C_1 \cos \frac{\pi t}{2} + C_2 \text{ sen } \frac{\pi t}{2}) + \frac{4^t}{19}$.

(d) $y_{t+2} + 5y_{t+1} + 2y_t = e^t$.

Solución homogénea:

$y_t = r^t$.

$r^{t+2} + 5r^{t+1} + 2r^t = 0$
$r^t \; (r^2 + 5r + 2) = 0$.

$r_1, r_2 = \frac{-5 \pm \sqrt{5^2 - 4*1*2}}{2*1}$
$r_1, r_2 = \frac{-5 \pm \sqrt{25-8}}{2}$
$r_1, r_2 = \frac{-5 \pm \sqrt{17}}{2}$
$r_1 = \frac{-5+\sqrt{17}}{2}$.
$r_2 = \frac{-5-\sqrt{17}}{2}$.

$y_t^h = C_1 \; (\frac{-5+\sqrt{17}}{2})^t + C_2 \; (\frac{-5-\sqrt{17}}{2})^t$.

Solución particular:

$y_t^p = \text{k}e^t$.

$y_{t+1} = \text{k } e^{t+1}$
$y_{t+1} = \text{ek } e^t$.

$y_{t+2} = \text{k } e^{t+2}$
$y_{t+2} = e^2\text{k } e^t$.

$e^2\text{k } e^t + 5\text{ek } e^t + 2\text{k } e^t = e^t$
$(e^2 + 5e + 2) \text{ k}e^t = e^t$
$(e^2 + 5e + 2) \text{ k} = \frac{e^t}{e^t}$
$(e^2 + 5e + 2) \text{ k} = 1$
$\text{k} = \frac{1}{e^2+5e+2}$.

$y_t^p = \frac{e^t}{e^2+5e+2}$.

Solución general:

$y_t = y_t^h + y_t^p$
$y_t = C_1\ (\frac{-5+\sqrt{17}}{2})^t + C_2\ (\frac{-5-\sqrt{17}}{2})^t + \frac{e^t}{e^2+5e+2}$.

(e) y_{t+2} - $2y_{t+1}$ + $5y_t = t$.

Solución homogénea:

$y_t = r^t$.

r^{t+2} - $2r^{t+1}$ + $5r^t = 0$
$r^t\ (r^2$ - 2r + 5)= 0.

$r_1, r_2 = \frac{-(-2)\pm\sqrt{(-2)^2-4*1*5}}{2*1}$
$r_1, r_2 = \frac{2\pm\sqrt{4-20}}{2}$
$r_1, r_2 = \frac{2\pm\sqrt{-16}}{2}$
$r_1, r_2 = \frac{2\pm 4i}{2}$
$r_1 = \frac{2+4i}{2}$= 1 + 2i.
$r_2 = \frac{2-4i}{2}$= 1 - 2i.

$\rho = \sqrt{1^2+2^2}$
$\rho = \sqrt{1+4}$
$\rho = \sqrt{5}$.

tan $\theta = \frac{2}{1}$
tan θ= 2.

$r_1, r_2 = \sqrt{5}$ [cos (arctan 2) $\pm$ i sen (arctan 2)].

$y_t^h = (\sqrt{5})^t$ [C_1 cos (t arctan 2) + C_2 sen (t arctan 2)].

Solución particular:

$y_t^p = k_1$t + k_2.

$y_{t+1} = k_1$ (t + 1) + k_2
$y_{t+1} = k_1$t + k_1 + k_2.

$y_{t+2} = k_1$ (t + 2) + k_2

$y_{t+2}= k_1 t + 2k_1 + k_2$.

$k_1 t + 2k_1 + k_2 - 2 (k_1 t + k_1 + k_2) + 5 (k_1 t + k_2)= t$
$4k_1 t + 4k_2= t$.

$4k_1= t$
$k_1= \frac{1}{4} t$.

$4k_2= 0$
$k_2= \frac{0}{4}$
$k_2= 0$.

$y_t^p= \frac{1}{4} t$.

<u>Solución general:</u>

$y_t= y_t^h + y_t^p$
$y_t= (\sqrt{5})^t [C_1 \cos (t \arctan 2) + C_2 \operatorname{sen} (t \arctan 2)] + \frac{1}{4} t$.

(f) $y_{t+2} - 2y_{t+1} + 5y_t = 4 + 2t$.

<u>Solución homogénea:</u>

$y_t= r^t$.

$r^{t+2} - 2r^{t+1} + 5r^t= 0$
$r^t (r^2 - 2r + 5)= 0$.

$r_1, r_2= \frac{-(-2)\pm\sqrt{(-2)^2-4*1*5}}{2*1}$
$r_1, r_2= \frac{2\pm\sqrt{4-20}}{2}$
$r_1, r_2= \frac{2\pm\sqrt{-16}}{2}$
$r_1, r_2= \frac{2\pm 4i}{2}$
$r_1= \frac{2+4i}{2}= 1 + 2i$.
$r_2= \frac{2-4i}{2}= 1 - 2i$.

$\rho= \sqrt{1^2 + 2^2}$
$\rho= \sqrt{1 + 4}$
$\rho= \sqrt{5}$.

$\tan \theta= \frac{2}{1}$
$\tan \theta= 2$.

r_1, r_2= $\sqrt{5}$ [cos (arctan 2) $\pm$ i sen (arctan 2)].

y_t^h= $(\sqrt{5})^t$ [C_1 cos (t arctan 2) + C_2 sen (t arctan 2)].

Solución particular:

y_t^p= k_1t + k_2.

y_{t+1}= k_1 (t + 1) + k_2
y_{t+1}= k_1t + k_1 + k_2.

y_{t+2}= k_1 (t + 2) + k_2
y_{t+2}= k_1t + 2k_1 + k_2.

k_1t + 2k_1 + k_2 - 2 (k_1t + k_1 + k_2) + 5 (k_1t + k_2)= 4 + 2t
4k_1t + 4k_2= 4 + 2t.

4k_1= 2t
k_1= $\frac{2}{4}$ t
k_1= $\frac{1}{2}$ t

4k_2= 4
k_2= $\frac{4}{4}$
k_2= 1.

y_t^p= $\frac{1}{2}$ t + 1.

Solución general:

y_t= y_t^h + y_t^p
y_t= $(\sqrt{5})^t$ [C_1 cos (t arctan 2) + C_2 sen (t arctan 2)] + $\frac{1}{2}$ t + 1.

(g) y_{t+2} + 5y_{t+1} + 2y_t= 18 + 6t + 8t^2.

Solución homogénea:

y_t= r^t.

r^{t+2} + 5r^{t+1} + 2r^t= 0
r^t (r^2 + 5r + 2)= 0.

r_1, r_2= $\frac{-5\pm\sqrt{5^2-4*1*2}}{2*1}$
r_1, r_2= $\frac{-5\pm\sqrt{25-8}}{2}$

$r_1, r_2= \frac{-5\pm\sqrt{17}}{2}$
$r_1= \frac{-5+\sqrt{17}}{2}$.
$r_2= \frac{-5-\sqrt{17}}{2}$.

$y_t^h= C_1\ (\frac{-5+\sqrt{17}}{2})^t + C_2\ (\frac{-5-\sqrt{17}}{2})^t$.

Solución particular:

$y_t^p= k_1 t^2 + k_2 \text{t} + k_3$.

$y_{t+1}= k_1\ (t+1)^2\ k_2\ (\text{t} + 1) + k_3$
$y_{t+1}= k_1\ (t^2 + 2\text{t} + 1) + k_2\text{t} + k_2 + k_3$
$y_{t+1}= k_1 t^2 + 2\text{t}\ k_1 + k_1 + k_2\text{t} + k_2 + k_3$
$y_{t+1}= k_1 t^2 + (2k_1 + k_2)\ \text{t} + (k_1 + k_2 + k_3)$.

$y_{t+2}= k_1\ (t+2)^2\ k_2\ (\text{t} + 2) + k_3$
$y_{t+2}= k_1\ (t^2 + 4\text{t} + 4) + k_2\text{t} + 2k_2 + k_3$
$y_{t+2}= k_1 t^2 + 4\text{t}\ k_1 + 4k_1 + k_2\text{t} + 2k_2 + k_3$
$y_{t+2}= k_1 t^2 + (4k_1 + k_2)\ \text{t} + (4k_1 + 2k_2 + k_3)$.

$k_1 t^2 + (4k_1 + k_2)\ \text{t} + (4k_1 + 2k_2 + k_3) + 5\ [k_1 t^2 + (2k_1 + k_2)\ \text{t} + (k_1 + k_2 + k_3)] + 2\ (k_1 t^2 + k_2\text{t} + k_3)= 18 + 6\text{t} + 8t^2$
$k_1 t^2 + (4k_1 + k_2)\ \text{t} + (4k_1 + 2k_2 + k_3) + 5k_1 t^2 + 5\ (2k_1 + k_2)\ \text{t} + 5\ (k_1 + k_2 + k_3) + 2k_1 t^2 + 2k_2\text{t} + 2k_3= 18 + 6\text{t} + 8t^2$
$k_1 t^2 + (4k_1 + k_2)\ \text{t} + (4k_1 + 2k_2 + k_3) + 5k_1 t^2 + (10k_1 + 5k_2)\ \text{t} + (5k_1 + 5k_2 + 5k_3) + 2k_1 t^2 + 2k_2\text{t} + 2k_3= 18 + 6\text{t} + 8t^2$
$8k_1 t^2 + (14k_1 + 8k_2)\ \text{t} + (9k_1 + 7k_2 + 8k_3)= 18 + 6\text{t} + 8t^2$.

$8k_1= 8$
$k_1= \frac{8}{8}$
$k_1= 1$.

$14k_1 + 8k_2= 6$
$14 * 1 + 8k_2= 6$
$14 + 8k_2= 6$
$8k_2= 6 - 14$
$8k_2= -8$
$k_2= \frac{-8}{8}$
$k_2= -1$.

$9k_1 + 7k_2 + 8k_3= 18$
$9 * 1 + 7\ (-1) + 8k_3= 18$
$9 - 7 + 8k_3= 18$
$2 + 8k_3= 18$
$8k_3= 18 - 2$
$8k_3= 16$

$k_3 = \frac{16}{8}$
$k_3 = 2.$

$y_t^p = t^2$ - t + 2.

Solución general:

$y_t = y_t^h + y_t^p$
$y_t = C_1\ (\frac{-5+\sqrt{17}}{2})^t + C_2\ (\frac{-5-\sqrt{17}}{2})^t + t^2$ - t + 2.

(h) $y_{t+2} + 5y_{t+1} + 2y_t = e^t + 18 + 6t + 8t^2$.

Solución homogénea:

$y_t = r^t$.

$r^{t+2} + 5r^{t+1} + 2r^t = 0$
$r^t\ (r^2 + 5r + 2) = 0.$

$r_1, r_2 = \frac{-5 \pm \sqrt{5^2 - 4*1*2}}{2*1}$
$r_1, r_2 = \frac{-5 \pm \sqrt{25-8}}{2}$
$r_1, r_2 = \frac{-5 \pm \sqrt{17}}{2}$
$r_1 = \frac{-5+\sqrt{17}}{2}$.
$r_2 = \frac{-5-\sqrt{17}}{2}$.

$y_t^h = C_1\ (\frac{-5+\sqrt{17}}{2})^t + C_2\ (\frac{-5-\sqrt{17}}{2})^t$.

Solución particular:

$y_t^p = y_t^{p_1} + y_t^{p_2}$
$y_t^p = e^t + k_1 t^2 + k_2 t + k_3$.

$y_t^{p_1} = ke^t$.

$y_{t+1} = k\ e^{t+1}$
$y_{t+1} = ek\ e^t$.

$y_{t+2} = k\ e^{t+2}$
$y_{t+2} = e^2 k\ e^t$.

$e^2 k\ e^t + 5ek\ e^t + 2k\ e^t = e^t$
$(e^2 + 5e + 2)\ ke^t = e^t$

$(e^2$ + 5e + 2) k= $\frac{e^t}{e^t}$
$(e^2$ + 5e + 2) k= 1
k= $\frac{1}{e^2+5e+2}$.

$y_t^{p_1}$= $\frac{e^t}{e^2+5e+2}$.

$y_t^{p_2}$= $k_1 t^2$ + k_2t + k_3.

y_{t+1}= k_1 $(t+1)^2$ k_2 (t + 1) + k_3
y_{t+1}= k_1 (t^2 + 2t + 1) + k_2t + k_2 + k_3
y_{t+1}= $k_1 t^2$ + 2t k_1 + k_1 + k_2t + k_2 + k_3
y_{t+1}= $k_1 t^2$ + (2k_1 + k_2) t + (k_1 + k_2 + k_3).

y_{t+2}= k_1 $(t+2)^2$ k_2 (t + 2) + k_3
y_{t+2}= k_1 (t^2 + 4t + 4) + k_2t + 2k_2 + k_3
y_{t+2}= $k_1 t^2$ + 4t k_1 + 4k_1 + k_2t + 2k_2 + k_3
y_{t+2}= $k_1 t^2$ + (4k_1 + k_2) t + (4k_1 + 2k_2 + k_3).

$k_1 t^2$ + (4k_1 + k_2) t + (4k_1 + 2k_2 + k_3) + 5 [$k_1 t^2$ + (2k_1 + k_2) t + (k_1 + k_2 + k_3)] + 2 ($k_1 t^2$ + k_2t + k_3)= 18 + 6t + 8t^2
$k_1 t^2$ + (4k_1 + k_2) t + (4k_1 + 2k_2 + k_3) + 5$k_1 t^2$ + 5 (2k_1 + k_2) t + 5 (k_1 + k_2 + k_3) + 2$k_1 t^2$ + 2k_2t + 2k_3= 18 + 6t + 8t^2
$k_1 t^2$ + (4k_1 + k_2) t + (4k_1 + 2k_2 + k_3) + 5$k_1 t^2$ + (10k_1 + 5k_2) t + (5k_1 + 5k_2 + 5k_3) + 2$k_1 t^2$ + 2k_2t + 2k_3= 18 + 6t + 8t^2
8$k_1 t^2$ + (14k_1 + 8k_2) t + (9k_1 + 7k_2 + 8k_3)= 18 + 6t + 8t^2.

8k_1= 8
k_1= $\frac{8}{8}$
k_1= 1.

14k_1 + 8k_2= 6
14 * 1 + 8k_2= 6
14 + 8k_2= 6
8k_2= 6 - 14
8k_2= -8
k_2= $\frac{-8}{8}$
k_2= -1.

9k_1 + 7k_2 + 8k_3= 18
9 * 1 + 7 (-1) + 8k_3= 18
9 - 7 + 8k_3= 18
2 + 8k_3= 18
8k_3= 18 - 2
8k_3= 16
k_3= $\frac{16}{8}$
k_3= 2.

$y_t^{p_2}= t^2$ - t + 2.

$y_t^{p}= \frac{e^t}{e^2+5e+2} + t^2$ - t + 2.

<u>Solución general:</u>

$y_t= y_t^h + y_t^p$
$y_t= C_1 \left(\frac{-5+\sqrt{17}}{2}\right)^t + C_2 \left(\frac{-5-\sqrt{17}}{2}\right)^t + \frac{e^t}{e^2+5e+2} + t^2$ - t + 2.

Ejercicio 7 (*).

Resolver la siguiente ecuación en diferencia de orden 2:

y_{t+2} - $3y_{t+1}$ - $4y_t = 2^t + t^3$.

Se parte de saber que la solución general de una ecuación en diferencias son las soluciones de la ecuación homogénea asociada a ésta más la solución particular:

$y_t = y_t^h + y_t^p$.

Siendo así, en primer lugar, se buscan las soluciones de la ecuación homogénea asociada a la ecuación en diferencias:

y_{t+2} - $3y_{t+1}$ - $4y_t = 0$.

Sea $y_t = r^t$, se tiene:

r^{t+2} - $3r^{t+1}$ - $4r^t = 0$
r^t (r^2 - 3r - 4)= 0.

Resolviendo mediante el método de Bhaskara, se hallan las soluciones de la ecuación homogénea:

$r_1, r_2 = \frac{-(-3) \pm \sqrt{(-3)^2 - 4*1(-4)}}{2*1}$
$r_1, r_2 = \frac{3 \pm \sqrt{9+16}}{2}$
$r_1, r_2 = \frac{3 \pm \sqrt{25}}{2}$
$r_1, r_2 = \frac{3 \pm 5}{2}$
$r_1 = \frac{3+5}{2} = \frac{8}{2} = 4$.
$r_2 = \frac{3-5}{2} = \frac{-2}{2} = -1$.

Entonces, la solución homogénea es la siguiente:

$y_t^h = C_1\ 4^t + C_2\ (-1)^t$.

En segundo lugar, dado que 1 no es raíz de la ecuación r^2 - 3r - 4= 0, se propone la siguiente solución particular:

$y_t^p = k_1 2^t + k_2 t^3 + k_3 t^2 + k_4 t + k_5$.

Resolviendo en la ecuación en diferencias, se tiene:

$[k_1 2^{t+2} + k_2\ (t+2)^3 + k_3\ (t+2)^2 + k_4\ (t+2) + k_5]$ -
$3\ [k_1 2^{t+1} + k_2\ (t+1)^3 + k_3\ (t+1)^2 + k_4\ (t+1) + k_5]$ -
$4\ (k_1 2^t + k_2 t^3 + k_3 t^2 + k_4 t + k_5) = 2^t + t^3$

$[k_1 2^{t+2} + k_2\ (t^3 + 6t^2 + 12t + 8) + k_3\ (t^2 + 4t + 4) + k_4 t + 2k_4 + k_5]$ -
3 $[k_1 2^{t+1} + k_2\ (t^3 + 3t^2 + 3t + 1) + k_3\ (t^2 + 2t + 1) + k_4 t + k_4 + k_5]$ -
$(4k_1 2^t + 4k_2 t^3 + 4k_3 t^2 + 4k_4 t + 4k_5) = 2^t + t^3$

$k_1 2^{t+2} + k_2 t^3 + 6k_2 t^2 + 12k_2 t + 8k_2 + k_3 t^2 + 4k_3 t + 4k_3 + k_4 t + 2k_4 + k_5$ -
3 $[k_1 2^{t+1} + k_2 t^3 + 3k_2 t^2 + 3k_2 t + k_2 + k_3 t^2 + 2k_3 t + k_3 + k_4 t + k_4 + k_5]$ -
$4k_1 2^t - 4k_2 t^3 - 4k_3 t^2 - 4k_4 t - 4k_5 = 2^t + t^3$

$k_1 2^{t+2} + k_2 t^3 + 6k_2 t^2 + 12k_2 t + 8k_2 + k_3 t^2 + 4k_3 t + 4k_3 + k_4 t + 2k_4 + k_5$ -
$(3k_1 2^{t+1} + 3k_2 t^3 + 9k_2 t^2 + 9k_2 t + 3k_2 + 3k_3 t^2 + 6k_3 t + 3k_3 + 3k_4 t + 3k_4 + 3k_5)$ -
$4k_1 2^t - 4k_2 t^3 - 4k_3 t^2 - 4k_4 t - 4k_5 = 2^t + t^3$

$k_1 2^{t+2} + k_2 t^3 + 6k_2 t^2 + 12k_2 t + 8k_2 + k_3 t^2 + 4k_3 t + 4k_3 + k_4 t + 2k_4 + k_5$ -
$3k_1 2^{t+1} - 3k_2 t^3 - 9k_2 t^2 - 9k_2 t - 3k_2 - 3k_3 t^2 - 6k_3 t - 3k_3 - 3k_4 t - 3k_4 - 3k_5$ -
$4k_1 2^t - 4k_2 t^3 - 4k_3 t^2 - 4k_4 t - 4k_5 = 2^t + t^3$

$-6k_1 2^t - 6k_2 t^3 - (3k_2 + 6k_3)\ t^2 + (3k_2 - 2k_3 - 6k_4)\ t + (5k_2 + k_3 - k_4 - 6k_5) = 2^t + t^3$

Luego, se tiene que las constantes k_1, k_2, k_3, k_4 y k_5, respectivamente, son iguales a:

$-6k_1 2^t = 2^t$
$k_1 = \frac{2^t}{-6*2^t}$
$k_1 = \frac{-1}{6}$.

$-6k_2 t^3 = t^3$
$k_2 = \frac{t^3}{-6t^3}$
$k_2 = \frac{-1}{6}$.

$-(3k_2 + 6k_3)\ t^2 = 0$
$3k_2 + 6k_3 = \frac{0}{-t^2}$
$3k_2 + 6k_3 = 0$
$6k_3 = -3k_2$
$k_3 = \frac{-3k_2}{6}$
$k_3 = \frac{-1}{2}\ k_2$
$k_3 = \frac{-1}{2}\ (\frac{-1}{6})$
$k_3 = \frac{1}{12}$.

$(3k_2 - 2k_3 - 6k_4)\ t = 0$
$3k_2 - 2k_3 - 6k_4 = \frac{0}{t}$
$3k_2 - 2k_3 - 6k_4 = 0$
$6k_4 = 3k_2 - 2k_3$
$k_4 = \frac{3k_2 - 2k_3}{6}$
$k_4 = \frac{1}{2}\ k_2 - \frac{1}{3}\ k_3$
$k_4 = \frac{1}{2}\ (\frac{-1}{6}) - \frac{1}{3}\ \frac{1}{12}$
$k_4 = \frac{-1}{12} - \frac{1}{36}$

$k_4 = \frac{-1}{9}$.

$5k_2 + k_3 - k_4 - 6k_5 = 0$
$6k_5 = 5k_2 + k_3 - k_4$
$k_5 = \frac{5k_2 + k_3 - k_4}{6}$
$k_5 = \frac{5}{6} k_2 + \frac{1}{6} k_3 - \frac{1}{6} k_4$
$k_5 = \frac{5}{6} (\frac{-1}{6}) + \frac{1}{6} \frac{1}{12} - \frac{1}{6} (\frac{-1}{9})$
$k_5 = \frac{-5}{36} + \frac{1}{72} + \frac{1}{54}$
$k_5 = \frac{-23}{216}$.

Entonces, la solución particular es la siguiente:

$y_t^p = \frac{-1}{6} 2^t - \frac{1}{6} t^3 + \frac{1}{12} t^2 - \frac{1}{9} t - \frac{23}{216}$.

Por lo tanto, la solución general de la ecuación en diferencias de orden 2 es:

$y_t = C_1 4^t + C_2 (-1)^t - \frac{1}{6} 2^t - \frac{1}{6} t^3 + \frac{1}{12} t^2 - \frac{1}{9} t - \frac{23}{216}$.